



## Phase II study of doxorubicin and thalidomide in patients with refractory aggressive fibromatosis

Xin Liu<sup>1,2</sup> · Huijie Wang<sup>1,2</sup> · Xianghua Wu<sup>1,2</sup> · Xiaonan Hong<sup>1,2</sup> · Zhiguo Luo<sup>1,2</sup> 

Received: 7 October 2017 / Accepted: 10 November 2017  
© Springer Science+Business Media, LLC, part of Springer Nature 2017

**Summary** *Background* To evaluate the efficacy and safety of doxorubicin (ADM) combined with thalidomide (THA) as a first-line treatment for patients with refractory aggressive fibromatosis (AF). *Patients and Methods* Eligible patients were treated with ADM 30 mg/m<sup>2</sup> on days 1–2 and THA 200 mg nightly on days 1–21 every 3 weeks for a maximum of six cycles. THA was then continued for a total of 1 year. The primary end point was response rate (RR). *Results* Fifteen patients were enrolled in the study. No patient had a complete response, but five patients had partial responses, resulting in a RR of 33%. Eight patients (53%) had stable disease and two patients (13%) had progressive disease, and the disease control rate was 87%. The median progression free survival (mPFS) was 20.6 months (95% confidence interval, 14.5–26.7 months). Patients with below normal baseline serum albumin levels had significantly inferior mPFS compared with those with normal baseline serum albumin (1.4 months versus 23.7 months,  $P = 0.045$ ). Grade 3/4 toxicities included leukopenia (33%), neutropenia (60%), febrile neutropenia (7%), nausea (7%), and vomiting (6.6%). *Conclusions* ADM plus THA was well-tolerated and effective as a first-line treatment for patients with refractory AF. However, patients with hypoalbuminemia at baseline had inferior clinical outcomes, and further studies are needed to investigate this issue.

**Keywords** Aggressive fibromatosis · Doxorubicin · Thalidomide · Hypoalbuminemia · Beta-catenin

### Introduction

Aggressive fibromatosis (AF, also known as desmoid tumor) is a histologically benign fibrous tissue tumor first described by Farlane in 1832. It is a rare disease, with an estimated annual incidence rate of only 2–4 cases per million. The etiology of AF remains unclear, though risk factors include high estrogen states (such as pregnancy), prior trauma, and familial adenomatous polyposis (FAP) [1–3].

AF can occur throughout the body and can be classified into intra-abdominal and extra-abdominal tumors. The most common tumor sites are the extremities, trunk wall, mesentery-omentum, and head and neck. AF does not metastasize, but manifests as progressive local infiltration and invasion of the muscle and surrounding tissue, which can cause pain,

deformity, functional disability, and even death if vital organs are involved [4].

Surgical resection with a negative margin is currently the main treatment for AF, though local recurrence rates remain as high as 30%–40%, even after wide local excision. Radiation therapy can be given when surgical margins are involved or if surgical resection is difficult, while systemic therapy can improve clinical outcomes in patients with refractory AF unsuitable for surgery or radiotherapy [5].

Doxorubicin (ADM) is the most active single agent for the treatment of soft tissue sarcomas, and ADM-based regimens are frequently used for the treatment of AF [6–8]. Thalidomide (THA) is an oral agent, first marketed as a sedative and subsequently withdrawn from the market following reports of teratogenicity in 1961 [9]. However, THA later demonstrated anti-angiogenic activities, through downregulation of several angiogenic and growth-promoting factors [10]. THA was approved for the treatment of multiple myeloma in 2006, and some studies have investigated its use in patients with Kaposi's sarcoma [11] and leiomyosarcoma (LMS) [12, 13]. However, there have been no reports of the use of THA in patients with AF.

We therefore conducted a phase II trial to evaluate the efficacy and safety of ADM in combination with THA (AT regimen) as a first-line treatment in patients with refractory AF.

✉ Zhiguo Luo  
luozhiguo88@163.com

<sup>1</sup> Department of Medical Oncology, Fudan University Shanghai Cancer Center, 270 Dong-An Road, Shanghai 200032, China

<sup>2</sup> Department of Oncology, Shanghai Medical College, Fudan University, Shanghai 200032, China

## Methods

### Patient eligibility

This single-arm, single-center, phase II study included patients who met the following eligibility criteria: age 18–75 years; histologically confirmed refractory AF; unresectable disease (defined as tumor >10 cm with unclear boundary, involved with joint or large vessels, recurrent tumor not suitable for surgery); at least one measurable lesion; no prior chemotherapy or THA; Eastern Cooperative Oncology Group (ECOG) performance status (PS) of 0 or 1; left ventricular ejection fraction (LVEF) >50%; adequate organ function (bone marrow function: white cell count  $\geq 3.5 \times 10^9/\text{L}$ , neutrophil count  $\geq 1.5 \times 10^9/\text{L}$ , platelet count  $\geq 80 \times 10^9/\text{L}$ , hemoglobin  $\geq 80 \text{ g/L}$ ; liver function: serum bilirubin  $\leq 1.5 \times$  upper limit of normal [ULN] and serum transaminase levels  $\leq 2.5 \times$  ULN; renal function: serum creatinine  $\leq 1.25 \times$  ULN and creatinine clearance rate  $\geq 50 \text{ mL/min}$ ). The exclusion criteria were as follows: pregnant or breast-feeding or child-bearing potential without using adequate contraception; any other current or prior malignancy (except excised cervical carcinoma in situ or squamous cell skin carcinoma treated by surgery only); and severe or uncontrolled medical conditions.

This study was approved by the Fudan University Shanghai Cancer Center Institutional Review Board and conducted according to the Declaration of Helsinki. All patients provided written informed consent prior to participation in this study.

### Treatment and assessment

ADM was administered at a dose of  $30 \text{ mg/m}^2$  on days 1–2, followed by oral THA 200 mg nightly on days 1–21, repeated every 3 weeks. AT was administered for a maximum of six cycles, and THA was continued for a total of 1 year. Treatment was discontinued in the event of progressive disease (PD), unacceptable toxic effects, or withdrawal of patient consent. Dose reductions of ADM were applied in cases of febrile neutropenia or grade 4 myelosuppression. The THA dose was reduced to 150 or 100 mg/day when THA-associated grade 3/4 toxic effects occurred for the first or second time, respectively. THA was stopped if associated grade 3/4 toxic effects occurred for a third time.

Response evaluation was performed according to the Response Evaluation Criteria in Solid Tumors (RECIST 1.0) every two cycles during the treatment period and every 3 months after treatment discontinuation. Complete response (CR) or partial response (PR) were confirmed by computed tomography (CT) scans performed at least 4 weeks apart. The

duration of response was measured from the time that the measurement criteria for CR/PR (whichever was first recorded) were first met until the date that recurrent or progressive disease was first objectively documented. Adverse events (AEs) were evaluated according to the National Cancer Institute Common Terminology Criteria for Adverse Events (version 3.0).

### Statistical analysis

The primary endpoint was response rate (RR). The secondary endpoints included progression-free survival (PFS), overall survival (OS), and AEs. Survival curves were generated using the Kaplan–Meier method and PFS was compared between groups using log-rank tests. PFS was calculated from the day of the first infusion to the date of documented disease progression or death from any cause. Patients who had not progressed at the time of the final analysis were censored at the date of their last tumor assessment. OS was calculated from the day of the first infusion to death. Patients alive at the final survival analysis were censored using the last contact date. Statistical analyses were performed using SPSS software (version 12.0; SPSS, Chicago, IL, USA). Survival and safety analyses were performed for all the patients.  $P < 0.05$  was considered significant.

## Results

### Patient disposition

A total of 15 patients with refractory AF were enrolled into the study in our center between June 2012 and April 2015. All the patients had discontinued treatment by the time of data cut-off at the end of May 2017.

### Patient characteristics

The baseline characteristics of the patients are listed in Table 1. The 15 patients included four men (27%) and 11 women (73%), with a median age of 41 years (range 24–63). All treated patients had an ECOG PS of 0 or 1 (PS 0: 33%; PS 1: 67%). The primary tumor site was located in the head and neck in one patient (7%), thorax in three patients (20%), abdomen in eight patients (53%), and in the extremities in three patients (20%). Ten patients (67%) had undergone prior surgery and eight patients (53%) had experienced at least two surgeries. Six patients (40%) had received prior radiotherapy, three patients (20%) had received prior systemic therapy with tamoxifen, and 12 patients (80%) had no prior systemic therapy. Among all 15 patients, one patient (7%) had FAP, one patient had a past history of atrial fibrillation and one patient had rheumatic heart disease.

**Table 1** Patient Characteristics (*N* = 15 patients)

| Characteristic                   | Number of patients (%) |
|----------------------------------|------------------------|
| Median age (range), years        | 41 (24–63)             |
| Gender                           |                        |
| Male                             | 4 (27%)                |
| Female                           | 11 (73%)               |
| ECOG PS                          |                        |
| 0                                | 5 (33%)                |
| 1                                | 10 (67%)               |
| Primary tumor site               |                        |
| Head and Neck                    | 1(7%)                  |
| Thorax                           | 3(20%)                 |
| Abdomen                          | 8(53%)                 |
| Extremity                        | 3(20%)                 |
| Prior therapies                  |                        |
| Surgery                          | 10(67%)                |
| Radiation                        | 6 (40%)                |
| Reported prior genetic diagnosis |                        |
| FAP                              | 1(7%)                  |

Abbreviations: ECOG: Eastern Cooperative Oncology Group; PS: performance status; FAP, familial adenomatous polyposis

## Efficacy

All 15 patients were evaluable for response. The median tumor size at baseline was 124 mm (range, 63–389 mm). Of the 15 patients, five (33%) had tumors measuring <10 cm, seven (28%) had tumors measuring 10–20 cm, and three (50%) had tumors measuring >20 cm. The AT regimen was administered for a total of 74 cycles, with a median of six cycles (range, 2–6).

Five patients (5/15) had PR and none had CR, resulting in a RR of 33%. The median duration of response was 16.3 months (95% confidence interval [CI], 9.7–33.0 months). Among the five patients with PR, three achieved PR after four cycles of treatment, one after six cycles, and one after two cycles of treatment (median four cycles). The CT scans of one patient with PR, in whom further tumor regression occurred during THA monotherapy, are shown in Fig. 1. Stable disease (SD) was observed in eight (8/15) patients (53%), including seven cases of tumor shrinkage. PD was observed in two (2/15) patients (13%). The disease-control rate (CR + PR + SD) was 87% (Table 2). Eleven patients (11/15, 73%) experienced improvement of symptoms.

The median follow-up time was 26.1 months. The median PFS (mPFS) in all 15 patients was 20.6 months (95% CI, 14.5–26.7 months) (Fig. 2). Two patients remained free of disease progression for >48 months. At the time of analysis, four patients had died of disease, nine patients remained alive, and two patients were lost to follow up. The median OS was therefore not calculated.

## Univariate analysis of PFS

Univariate analysis identified hypoalbuminemia as the only factor predicting inferior PFS. The mPFS values were 23.7 months and 1.4 months in patients with normal ( $\geq 35$  g/L) and low ( $< 35$  g/L) baseline serum albumin levels, respectively ( $P = 0.045$ ) (Fig. 3). Other baseline characteristics such as gender, age, hemoglobin levels, tumor size and primary tumor sites were not associated with PFS in univariate analysis.

## Safety

All 15 patients were evaluated for safety. Treatment was generally well-tolerated and the main toxicities were hematological, with grade 3/4 leukopenia, neutropenia, and febrile neutropenia occurring in 33%, 60%, and 7% of patients, respectively. Non-hematological toxicities were moderate, with the most common grade 3/4 non-hematological toxicities being nausea (7%) and vomiting (6.6%). Other grade 1/2 non-hematological toxicities occurring in >20% of the patients included leukopenia (53%), neutropenia (27%), anemia (60%), alopecia (87%), anorexia (27%), nausea (27%), vomiting (40%), fatigue (33%), edema (20%), and sinus bradycardia (20%). There were no treatment-related deaths. Three patients experienced dose reduction of ADM because of grade 4 neutropenia or febrile neutropenia. THA treatment was well-tolerated in all patients, with no dose reduction. All grades of AEs are listed in Table 3.

LVEF was evaluated by echocardiography in all patients before treatment and after every two cycles while on trial. No decline in LVEF was observed in any patients. No grade 3/4 cardiac toxicity was experienced, even in the two patients with past histories of atrial fibrillation and rheumatic heart disease, respectively.

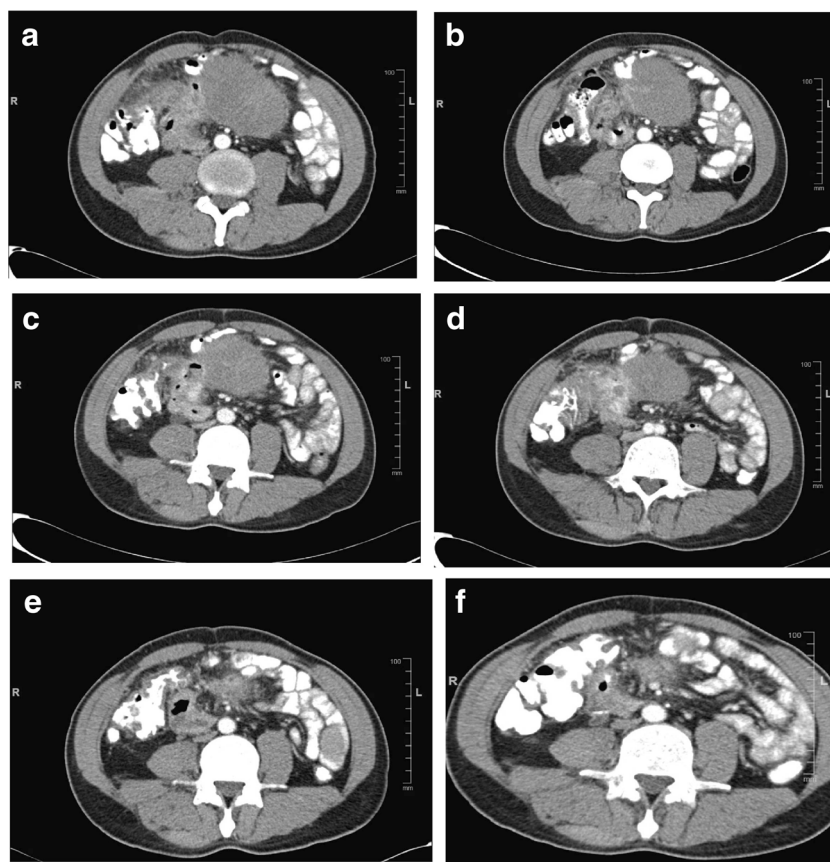
## Subsequent treatments

Ten patients experienced disease progression up to May 2017. Among subsequent treatments, two patients received radiotherapy, two underwent amputations, two received chemotherapy (methotrexate/vinblastine), and one patient received imatinib and pazopanib. Subsequent treatments were unknown in three patients.

## Discussion

In this study, an AT regimen provided effective first-line treatment for patients with refractory AF, resulting in a RR of 33%, disease-control rate of 87%, and mPFS of 20.6 months. Patients achieved PR after a median of four cycles of chemotherapy, suggesting that the tumor had a slow response to the

**Fig. 1** The computed tomographic scanings of 1 patient who achieved PR after AT chemotherapy. A showed pelvic tumor at baseline. B,C,D showed pelvic tumor regressed after 2, 4, 6 cycles of AT chemotherapy, respectively. E, F showed pelvic tumor further regressed during THA maintenance therapy. Abbreviations: AT, doxorubicin and thalidomide; PR, partial response; THA, thalidomide



therapy. Treatment was generally well-tolerated and the predominant grade 3/4 treatment-related toxic effects were leukopenia (33%), neutropenia (60%), febrile neutropenia (7%), nausea (7%), and vomiting (6.6%). The addition of THA to ADM was not associated with any significant increase in toxicities.

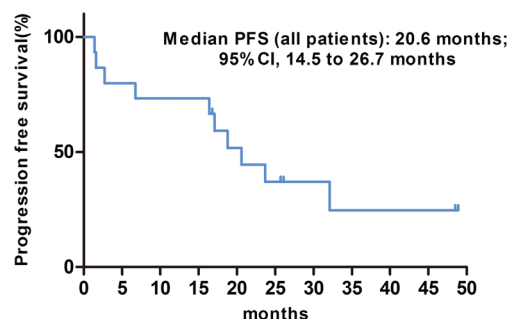
Current options for systemic therapy of AF include nonsteroidal anti-inflammatory drugs, hormonal agents (such as tamoxifen) [14], chemotherapy, and targeted agents (imatinib mesylate [15] and  $\gamma$ -secretase inhibitor PF-03084014 [16]). However, because of the rarity of the disease, only a few retrospective studies or phase II trials with small sample sizes

have explored chemotherapy regimens in patients with refractory AF, including weekly methotrexate/vinblastine and ADM-based regimens. Anthracycline-containing regimens were reported to have a higher RR in patients with AF. In a prospective study of seven patients with symptomatic unresectable FAP-associated AF treated with ADM plus dacarbazine (DTIC), Gega et al. reported a RR of 100% (7/7) and CR rate of 43% (3/7) [17]; however, the ADM-DTIC regimen was associated with serious toxicities, including myelosuppression and cardiotoxicity. Constantinidou et al. retrospectively analyzed 12 patients with AF who received pegylated liposomal ADM and found a RR of 36% [18].

**Table 2** Overall responses

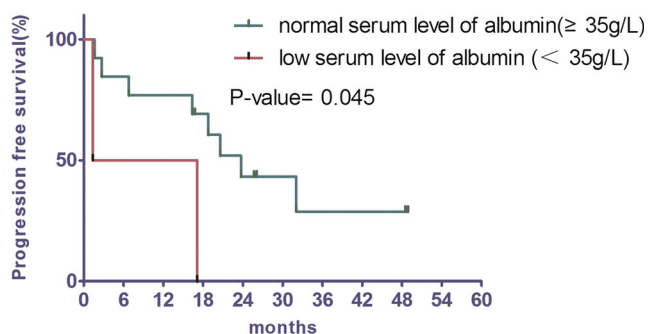
|                    | Number | Percentage (%) |
|--------------------|--------|----------------|
| All Patients       | 15     | 100%           |
| CR                 | 0      | 0%             |
| PR                 | 5      | 33%            |
| SD                 | 8      | 53%            |
| PD                 | 2      | 13%            |
| RR(CR + PR)        | 5      | 33%            |
| DCR (CR + PR + SD) | 13     | 87%            |

CR, complete response; PR, partial response; SD, stable disease; PD, progressive disease; RR, response rate; DCR, disease control rate



**Fig. 2** Kaplan–Meier estimates of progression free survival (PFS) among fifteen patients with refractory aggressive fibromatosis treated with doxorubicin and thalidomide





**Fig. 3** Kaplan–Meier estimates of progression free survival (PFS) among fifteen patients with refractory aggressive fibromatosis with normal and abnormal baseline serum albumin levels

Anti-angiogenic agents have been explored in patients with AF. Gounder et al. reported efficacy of sorafenib in AF, with a RR of 25% [19]. Grünwald et al. treated a patient with progressive AF after failure of chemotherapy with single-agent bevacizumab for 1 year, with symptomatic and radiologic response [20]. However, agents such as sorafenib and bevacizumab are expensive and have shown obvious toxicities, including hypertension, proteinuria, and diarrhea.

THA has been shown to exert anti-angiogenic effects through inhibiting angiogenic promoting factors and improving oxygenation of the tumor vascular bed, potentially sensitizing the tumor to radiotherapy [21]. Du et al. found that THA might inhibit tumor growth via cyclooxygenase-2 degradation independent of anti-angiogenesis [22]. Furthermore, THA has already demonstrated promising activity against multiple myeloma, Kaposi's sarcoma and myelofibrosis. Case reports and pilot studies have documented isolated responses to THA in some soft tissue sarcomas (STS), such as refractory uterine LMS, angiosarcoma, and osteosarcoma [23, 24]. McMeekin et al. launched a phase II trial of THA in 29 patients with refractory uterine LMS of whom 24% experienced SD [13]. THA in combination with temozolomide has also been evaluated in a phase II study in patients with unresectable or metastatic LMS, with a RR of 10%, and SD rate of 24% [12].

To the best of our knowledge the current study provides the first report of the use of THA in patients with AF. Tumor regression occurred during THA monotherapy, suggesting that THA might have an antitumor effect and might offer a new treatment option in patients with AF. Oral THA, administered on an outpatient basis, had only mild toxicities and allowed patients to have a good quality of life. Furthermore,

**Table 3** All Grades of Adverse Events (National Cancer Institute Common Toxicity Criteria, Version 3.0)

| Adverse Event (N = 15)             | Grade 1 | Grade 2 | Grade 3 | Grade 4 |
|------------------------------------|---------|---------|---------|---------|
| <b>Hematological toxicity</b>      |         |         |         |         |
| Leukopenia                         | —       | 8(53%)  | 3(20%)  | 2(13%)  |
| Neutropenia                        | —       | 4(27%)  | 5(33%)  | 4(27%)  |
| Febrile neutropenia                | —       | —       | 1(7%)   | —       |
| Anemia                             | 2(13%)  | 7(47%)  | —       | —       |
| <b>Non-hematological toxicity</b>  |         |         |         |         |
| Alopecia                           | 6(40%)  | 7(47%)  | —       | —       |
| Anorexia                           | 3(20%)  | 1(7%)   | —       | —       |
| Nausea                             | 4(27%)  | —       | 1(7%)   | —       |
| Vomiting                           | 5(33%)  | 1(7%)   | 1(7%)   | —       |
| Stomatitis                         | —       | 1(7%)   | —       | —       |
| Constipation                       | 2(13%)  | —       | —       | —       |
| Alanine aminotransferase increased | 2(13%)  | —       | —       | —       |
| Bilirubin increased                | 1(7%)   | —       | —       | —       |
| Hypoalbuminemia                    | —       | 2(13%)  | —       | —       |
| Fatigue                            | 5(33%)  | —       | —       | —       |
| Somnolence                         | 2(13%)  | —       | —       | —       |
| Rash                               | 1(7%)   | 1(7%)   | —       | —       |
| Urticaria                          | —       | 1(7%)   | —       | —       |
| Edema                              | 3(20%)  | —       | —       | —       |
| Dizziness                          | 1(7%)   | —       | —       | —       |
| Infection                          | —       | 2(13%)  | —       | —       |
| Fever                              | 2(13%)  | —       | —       | —       |
| Sinus bradycardia                  | 3(20%)  | —       | —       | —       |

Data are shown as No. (%)

THA is relatively cheap, and thus offers a good choice for maintenance therapy in patients with AF, especially for patients unable to afford expensive targeted agents.

The adenomatous polyposis coli (*APC*) and beta-catenin (*CTNNB1*) genes are involved in the pathogenesis of AF. FAP-driven AF is usually associated with inherited mutations in the *APC* tumor suppressor gene, whereas sporadic AF is usually associated with mutations in *CTNNB1*, with a mutation rate of *CTNNB1* in sporadic AF as high as 86% [25]. Mutations in either *CTNNB1* or *APC* may result in increased nuclear expression of beta-catenin, leading to aberrant expression of genes, including the anti-apoptotic *BCL2*. THA has been reported to inhibit the expression of *BCL2*. Marriott et al. found that a subclass of THA analogs exerted antitumor activity through altered expression of *Bcl-2* family proteins [26], while Liu et al. found a dramatic decrease in *BCL2* expression in multiple myeloma following THA. These results suggested that combination with THA might enhance the cytotoxic effects of other cytotoxic agents [27]. Zhu et al. found that THA caused apoptosis in osteosarcoma cells by decreasing the expression of *BCL2* [28]. THA may thus exert anti-tumor effects by promoting apoptosis via inhibition of *BCL2* gene expression. Further studies are needed to explore this possible mechanism of THA in AF.

The current results suggested that patients with low baseline serum albumin levels had significantly poorer mPFS than patients with normal baseline serum albumin levels. Hypoalbuminemia has been identified as a poor prognostic factor in STS. Barreto-Andrade et al. analyzed 61 primary adult STS, and found that hypoalbuminemia was independently associated with decreased OS (RR 5.0 [95% CI: 2.1–9.4]) [29]. Biswas et al. showed that serum albumin  $\leq 3.4$  g/dL ( $P < 0.001$ ) was an independent predictor of poor outcome in patients with metastatic Ewing's sarcoma family of tumors, and also found that patients with hypoalbuminemia had higher incidences of systemic symptoms and multiple organ metastases compared with those with normal albumin levels [30]. Panotopoulos et al. also demonstrated that high albumin was associated with improved overall and disease-specific survival in patients with liposarcoma [31]. The current study provides the first report of an association between serum albumin levels and AF; however, because of the small sample size, only two patients had low baseline serum albumin levels. Further studies are therefore needed to verify these preliminary results.

In conclusion, the results of the current study indicated that the AT regimen could offer an effective and well-tolerated option for the first-line treatment of patients with refractory AF. However, patients with baseline hypoalbuminemia may experience inferior clinical outcomes. In light of the relatively small sample size and single-arm design of the current study, further basic and clinical studies are needed to clarify the value of THA and the significance of albumin levels in patients with AF.

**Funding** There was no funding for this study.

**Compliance with ethical standards**

**Conflict of interest** The authors declare no conflict of interest.

**Ethical approval** All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

**Informed consent** Informed consent was obtained from all individual participants included in the study.

## References

1. Ferenc T, Sygut J, Kopczynski J, Mayer M, Latos-Bielenska A, Dziki A, Kulig A (2006) Aggressive fibromatosis (desmoid tumors): definition, occurrence, pathology, diagnostic problems, clinical behavior, genetic background. *Pol J Pathol Off J Pol Soc Pathologists* 57(1):5–15
2. Calvert GT, Monument MJ, Burt RW, Jones KB, Randall RL (2012) Extra-abdominal desmoid tumors associated with familial adenomatous polyposis. *Sarcoma* 2012:726537. <https://doi.org/10.1155/2012/726537>
3. Kasper B, Strobel P, Hohenberger P (2011) Desmoid tumors: clinical features and treatment options for advanced disease. *Oncologist* 16(5):682–693. <https://doi.org/10.1634/theoncologist.2010-0281>
4. Wong SL (2008) Diagnosis and management of desmoid tumors and fibrosarcoma. *J Surg Oncol* 97(6):554–558. <https://doi.org/10.1002/jso.20981>
5. de Camargo VP, Keohan ML, D'Adamo DR, Antonescu CR, Brennan MF, Singer S, Ahn LS, Maki RG (2010) Clinical outcomes of systemic therapy for patients with deep fibromatosis (desmoid tumor). *Cancer* 116(9):2258–2265. <https://doi.org/10.1002/cncr.25089>
6. Seiter K, Kemeny N (1993) Successful treatment of a desmoid tumor with doxorubicin. *Cancer* 71(7):2242–2244
7. Patel SR, Evans HL, Benjamin RS (1993) Combination chemotherapy in adult desmoid tumors. *Cancer* 72(11):3244–3247
8. Ezumi K, Yamamoto H, Takemasa I, Nomura M, Ikeda M, Sekimoto M, Monden M (2008) Dacarbazine-doxorubicin therapy ameliorated an extremely aggressive mesenteric desmoid tumor associated with familial adenomatous polyposis: report of a case. *Jpn J Clin Oncol* 38(3):222–226. <https://doi.org/10.1093/jjco/hyn004>
9. Yang TJ, Yang TS, Liang HM (1963) Thalidomide and congenital abnormalities. *Lancet* 1(7280):552–553
10. Vacca A, Scavelli C, Montefusco V, Di Pietro G, Neri A, Mattioli M, Biciato S, Nico B, Ribatti D, Dammacco F, Corradini P (2005) Thalidomide downregulates angiogenic genes in bone marrow endothelial cells of patients with active multiple myeloma. *J Clin Oncol Off J Am Soc Clin Oncol* 23(23):5334–5346. <https://doi.org/10.1200/JCO.2005.03.723>
11. Little RF, Wyvill KM, Pluda JM, Welles L, Marshall V, Figg WD, Newcomb FM, Tosato G, Feigal E, Steinberg SM, Whitby D, Goedert JJ, Yarchoan R (2000) Activity of thalidomide in AIDS-related Kaposi's sarcoma. *J Clin Oncol Off J Am Soc Clin Oncol* 18(13):2593–2602. <https://doi.org/10.1200/JCO.2000.18.13.2593>
12. Boyar MS, Hesdorffer M, Keohan ML, Jin Z, Taub RN (2008) Phase II study of Temozolomide and thalidomide in patients with Unresectable or metastatic Leiomyosarcoma. *Sarcoma* 2008: 412503. <https://doi.org/10.1155/2008/412503>

13. McMeekin DS, Sill MW, Darcy KM, Stearns-Kurosawa DJ, Webster K, Waggoner S, Benbrook D (2007) A phase II trial of thalidomide in patients with refractory leiomyosarcoma of the uterus and correlation with biomarkers of angiogenesis: a gynecologic oncology group study. *Gynecol Oncol* 106(3):596–603. <https://doi.org/10.1016/j.ygyno.2007.05.013>
14. Hansmann A, Adolph C, Vogel T, Unger A, Moeslein G (2004) High-dose tamoxifen and sulindac as first-line treatment for desmoid tumors. *Cancer* 100(3):612–620. <https://doi.org/10.1002/cncr.11937>
15. Chugh R, Wathen JK, Patel SR, Maki RG, Meyers PA, Schuetze SM, Priebat DA, Thomas DG, Jacobson JA, Samuels BL, Benjamin RS, Baker LH, Sarcoma Alliance for Research through C (2010) Efficacy of imatinib in aggressive fibromatosis: results of a phase II multicenter sarcoma alliance for research through collaboration (SARC) trial. *Clin Cancer Res: An Official Journal of the American Association for Cancer Research* 16 (19):4884–4891. doi:<https://doi.org/10.1158/1078-0432.CCR-10-1177>
16. Kummur S, O'Sullivan Coyne G, Do KT, Turkbey B, Meltzer PS, Polley E, Choyke PL, Meehan R, Vilimas R, Horneffer Y, Juwara L, Lih A, Choudhary A, Mitchell SA, Helman LJ, Doroshow JH, Chen AP (2017) Clinical activity of the gamma-secretase inhibitor PF-03084014 in adults with Desmoid tumors (aggressive fibromatosis). *J Clin Oncol Off J Am Soc Clin Oncol* 35(14):1561–1569. <https://doi.org/10.1200/JCO.2016.71.1994>
17. Gega M, Yanagi H, Yoshikawa R, Noda M, Ikeuchi H, Tsukamoto K, Oshima T, Fujiwara Y, Gondo N, Tamura K, Utsunomiya J, Hashimoto-Tamaoki T, Yamamura T (2006) Successful chemotherapeutic modality of doxorubicin plus dacarbazine for the treatment of desmoid tumors in association with familial adenomatous polyposis. *J Clin Oncol Off J Am Soc Clin Oncol* 24(1):102–105. <https://doi.org/10.1200/JCO.2005.02.1923>
18. Constantinidou A, Jones RL, Scurr M, Al-Muderis O, Judson I (2009) Pegylated liposomal doxorubicin, an effective, well-tolerated treatment for refractory aggressive fibromatosis. *Eur J Cancer* 45(17):2930–2934. <https://doi.org/10.1016/j.ejca.2009.08.016>
19. Gounder MM, Lefkowitz RA, Keohan ML, D'Adamo DR, Hameed M, Antonescu CR, Singer S, Stout K, Ahn L, Maki RG (2011) Activity of Sorafenib against desmoid tumor/deep fibromatosis. *Clin Cancer Res: An Official Journal of the American Association for Cancer Research* 17(12):4082–4090. <https://doi.org/10.1158/1078-0432.CCR-10-3322>
20. Grunwald V, Langer F, Raatschen HJ, Beilken A (2016) Proof of principle for bevacizumab activity in desmoid-type fibromatosis. *Clin Sarcoma Res* 6:5. <https://doi.org/10.1186/s13569-016-0045-3>
21. Ansiaux R, Baudalet C, Jordan BF, Beghein N, Sonveaux P, De Wever J, Martinive P, Gregoire V, Feron O, Gallez B (2005) Thalidomide radiosensitizes tumors through early changes in the tumor microenvironment. *Clin Cancer Res: An Official Journal of the American Association for Cancer Research* 11(2 Pt 1):743–750
22. GJ D, Lin HH, QT X, Wang MW (2005) Thalidomide inhibits growth of tumors through COX-2 degradation independent of antiangiogenesis. *Vasc Pharmacol* 43(2):112–119. <https://doi.org/10.1016/j.vph.2005.04.003>
23. Raina V, Sengar M, Shukla NK, Deo SS, Mohanty BK, Sharma D, Ray R, Das P, Rath GK (2007) Complete response from thalidomide in angiosarcoma after treatment of breast cancer. *J Clin Oncol Off J Am Soc Clin Oncol* 25(7):900–901. <https://doi.org/10.1200/JCO.2006.09.7261>
24. Tsai YC, CT W, Hong RL (2005) Response of refractory osteosarcoma to thalidomide and celecoxib. *Lancet Oncol* 6(12):997–999. [https://doi.org/10.1016/S1470-2045\(05\)70468-X](https://doi.org/10.1016/S1470-2045(05)70468-X)
25. van Broekhoven DL, Verhoef C, Grunhagen DJ, van Gorp JM, den Bakker MA, Hinrichs JW, de Voijs CM, van Dalen T (2015) Prognostic value of CTNNB1 gene mutation in primary sporadic aggressive fibromatosis. *Ann Surg Oncol* 22(5):1464–1470. <https://doi.org/10.1245/s10434-014-4156-x>
26. Marriott JB, Clarke IA, Czajka A, Dredge K, Childs K, Man HW, Schafer P, Govinda S, Muller GW, Stirling DI, Dalgleish AG (2003) A novel subclass of thalidomide analogue with anti-solid tumor activity in which caspase-dependent apoptosis is associated with altered expression of bcl-2 family proteins. *Cancer Res* 63(3):593–599
27. Liu WM, Strauss SJ, Chaplin T, Shahin S, Propper DJ, Young BD, Joel SP, Malpas JS (2004) S-thalidomide has a greater effect on apoptosis than angiogenesis in a multiple myeloma cell line. *Hematol J Off J Eur Haematol Assoc* 5(3):247–254. <https://doi.org/10.1038/sj.thj.6200351>
28. Zhu J, Yang Y, Liu S, Xu H, Wu Y, Zhang G, Wang Y, Wang Y, Liu Y, Guo Q (2016) Anticancer effect of thalidomide in vitro on human osteosarcoma cells. *Oncol Rep* 36(6):3545–3551. <https://doi.org/10.3892/or.2016.5158>
29. Barreto-Andrade JC, Medina-Franco H (2009) Serum albumin is an independent prognostic factor for survival in soft tissue sarcomas. *Rev Investig Clin Organo del Hospital de Enfermedades de la Nutricion* 61(3):198–204
30. Biswas B, Rastogi S, Khan SA, Shukla NK, Deo SV, Agarwala S, Sharma DN, Thulkar S, Vishnubhatla S, Pathania S, Bakhshi S (2014) Hypoalbuminaemia is an independent predictor of poor outcome in metastatic Ewing's sarcoma family of tumours: a single institutional experience of 150 cases treated with uniform chemotherapy protocol. *Clin Oncol* 26(11):722–729. <https://doi.org/10.1016/j.clon.2014.05.006>
31. Panotopoulos J, Posch F, Funovics PT, Willegger M, Scharrer A, Lamm W, Brodowicz T, Windhager R, Ay C (2016) Elevated serum creatinine and low albumin are associated with poor outcomes in patients with liposarcoma. *J Orthop Res: Official Publication of the Orthopaedic Research Society* 34(3):533–538. <https://doi.org/10.1002/jor.23002>